

PALAK RAWAT

+91-7417179776 | rawatpalak140@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Full-stack developer and final-year B.Tech CSE student with hands-on experience building role-based web applications using React, Node.js, Express.js, and MySQL. Projects go beyond tutorials — implementing real algorithms (A*, Dijkstra, CPU scheduling) and system-level designs. AWS Certified Solutions Architect with strong foundations in DSA, Operating Systems, and Compiler Design. Seeking a full-stack internship to contribute to real product development and grow in a fast-moving environment.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, JavaScript

Web Technologies: HTML, CSS, React.js, Node.js, Express.js, REST APIs

Databases: MySQL

Cloud & DevOps: AWS (Solutions Architect Certified), Git, Linux (Ubuntu), Visual Studio Code

Core CS Concepts: Data Structures & Algorithms, OOP, Operating Systems, CPU Scheduling, Compiler Design

PROJECTS

Learning Management System (LMS) | *React, Node.js, Express.js, MySQL*

- Built a full-stack LMS with three-tier role-based access (student, instructor, admin) using React, Node.js, Express.js, and MySQL — handling distinct permission logic for each user type.
- Designed and implemented RESTful APIs in Node.js/Express.js enabling real-time data sync across user roles with persistent storage in MySQL.
- Built responsive React.js interfaces with component-based architecture, ensuring consistent UX across device sizes and user roles.

Syntax Analyzer with Smart Suggestions | *Python, Compiler Design*

- Built a full compiler pipeline (Lexer → Parser → AST) for syntax analysis of a custom programming language.
- Implemented multi-error detection with precise diagnostics (line/column + expected tokens) and a smart suggestion engine to recommend fixes for common syntax errors.
- Developed context-aware autocomplete using parser state and symbol table; generated and visualized Abstract Syntax Trees (AST) using Graphviz.

Inventory Management & Delivery Optimization System | *C++, DSA*

- Optimized delivery routing using A* and Dijkstra's algorithms with priority queues and dynamic programming, reducing redundant route evaluations and improving request throughput.
- Applied dynamic programming to efficiently handle delivery requests and route selection across a graph-based inventory model.

Emotion-Aware Resource Manager for Operating System | *Linux, OS Concepts*

- Designed an emotion-aware resource management module to dynamically adjust CPU scheduling policies based on real-time user states.
- Implemented Shortest Job First (SJF), Round Robin, and Lottery Scheduling for adaptive resource allocation, enhancing system responsiveness.

EDUCATION

Graphic Era Hill University, Dehradun 2023 – 2027

B.Tech in Computer Science and Engineering | CGPA: 7.67/10

CERTIFICATIONS

- **AWS Cloud Quest: Solutions Architect** (2025)
- **AWS CloudFormation Stacks – Troubleshooting** (2025)
- **Google Cloud Computing Foundations – NPTEL** (2024)
- **Machine Learning Plan – AWS Skill Builder** (2026)